

Part list for handheld probe

- - 1 x Seeed Studio XIAO ESP32-S3 (8MB Flash, 8MB SPI-RAM) or ESP32-S3 Super Mini (4MB Flash, 2 MB SPI-RAM) with battery charging support.
(<https://www.amazon.de/dp/B0BYSB66S5>, <https://www.amazon.de/dp/B0FKH1F8VV>, <https://eckstein-shop.de/Seeed-Studio-XIAO-ESP32S3-Dual-Core-24GHz-Wi-Fi-BLE50-Battery-Charge-Supported>)
- - 1 x „Golden Morning Electronic“ GME 12864-11-12-13, 0,96", 27.42 x 27.85 mm, OLED SSD1306 I2C display module, 128 x 64 pixels twin-color, blue or white. With housing rev. C also a similar 0.96 Inch OLED display with a little bit different mounting holes and board sizes. Important: **Pin order must be "GND,VCC,SCL,SDA"**. The firmware supports SSD1306 and SH1106 display controllers, or compatible types, with 128 by 64 pixels visible resolution.
(<https://www.amazon.de/dp/B0BWJFVYZR>, <https://www.amazon.de/dp/B0D2P4J1R6>, <https://www.amazon.de/dp/B0F13SZ757>, <https://www.amazon.de/dp/B0F13QRNM3>)
- - 1 x SPST Button Switch, 6 mm Head, 7.2 mm thread, innner height 8.5 mm, outer height 12.8 mm or near.
(<https://www.amazon.de/dp/B088WQFBDR>)
- - 1 x MTS-101 mini toggle switch, ON/OFF.
(<https://www.amazon.de/dp/B07SBQ6TS3> or <https://www.amazon.de/dp/B07JPDMGX4>)
- - 1 x Keppower 14500 - 1100 mAh, 3,6-3,7V Li-Ion battery PCB protected P1450C3, or similar type.
(https://www.akkuteile.de/lithium-ionen-akkus/14500/keppower/keppower-14500-1100-mah-3-6-3-7v-li-ion-akku-pcb-geschuetzt-p1450c3_12073_2785)
- - 1 x Battery holder AA (14500), width 17 mm, length 60 mm, height 15 mm.
(<https://www.amazon.de/dp/B08Z32KMHN>)
- - 1 x Sensor module (supported types: SHT2x/HTU21d/BME280/BME280/HDC1080/ATH10/ATH21) with I2C interface. e.g. ready to use FS200, FS304 or FS400 modules.
(<https://www.amazon.de/dp/B0C6L73L8S>, <https://www.amazon.de/dp/B07X33M2DK>, <https://www.amazon.de/dp/B0FJHZ7387>, <https://www.amazon.de/dp/B0H83PVXF9>, <https://www.amazon.de/dp/B0DSVMSZ3L>, <https://www.amazon.de/dp/B0CLS3NB4Q>)
- - 1 x M4 press-fit nut.
(<https://www.amazon.de/dp/B0CRHK8P9P>)
- - 1 x M4 x 16 mm countersunk screw
(<https://www.amazon.de/GUEEYUAW-Senkkopfschrauben-Edelstahl-Senkkopf-Innensechskant/dp/B0FLQF2CG8>)

Note: There are existing also 0,96" OLED display boards with the **not usable** "**VCC,GND,SCL,SDA**" order. To connect the display directly on the top of the ESP32-S3 board (5V, GND, 3V, GPIO, GPIO), we need GND 1st and then VCC !